SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club Submission

Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Ctrl + Sleep
Selected Domain	Domain 2
Selected SDG Track (1–17)	8
Project / Solution Name	QuoteBot AI, Instant WhatsApp Quotation Extractor & PDF Report Generator
Name (Leader)	Chin Xian Jing
Email & Phone (Leader)	xianjing0702@gmail.com
University Affiliation(s)	University of Nottingham Malaysia

Number of Team Members 4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved. ***[Maximum 250 words]***

In emerging markets across Asia and Latin America, over 80% of B2B transactions rely on unstructured WhatsApp communications. Daily procurement involves millions of price inquiries, handwritten photo quotes, voice messages, and scattered PDF attachments exchanged across fast-moving chat groups.

This operational bottleneck forces small business owners, procurement officers, and vendors to spend up to 15 hours weekly manually copying pricing details and terms into spreadsheets. This manual workflow is highly error-prone, causing miscommunications, incorrect billing, and lost sales due to delayed response times.

The economic impact is significant. SMEs account for over 90% of global businesses and 50% of employment. Slow transaction processing suppresses cash flow and liquidity, leaving small vendors unable to compete with enterprises utilizing automated ERP systems. Furthermore, without digitized records, SMEs cannot build credit histories or access formal supply chain financing, trapping them in informality.

Our solution bridges chat conversations with structured business processes. By turning chaotic WhatsApp messages into clean, actionable data, we enable small businesses to digitize quotes, compare suppliers, and finalize deals in minutes instead of hours—without manual errors or delays. This levels the playing field, empowering businesses to thrive in the digital economy.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Solution Overview Quote Bot AI our WhatsApp bot is an intelligent in-line conversational assistant for clients that converts free-formed chat conversations into structured data to automate the quote lifecycle into optimized supplier pricing analysis, directly within WhatsApp. It applies generative AI through a multi-modal GenAI pipeline that employs multiple Vision LLMs (GPT-4o / Claude 3.5 Sonnet) for parsing unstructured chat histories, voice notes, handwritten receipts and PDF quotes. It enforces JSON Schema rule constraints on LLM output to reliably extract vendor metadata, line item quantities, unit prices, currencies, payment terms, and call-to-action byposing in-line interactive summarization, vendor comparison & procurement insights, and final PDF quote report generation. Technologies & skills we will apply that we learnt from SDC Phase 1 workshops: - GenAI APIs & Prompt Engineering to achieve structured function calling & machine-interpretation of free-format Chat history, voice notes, image to text conversion, with multi-turn conversation context to resolve noisy, human-generated NLP inputs - Database Architecture to create relational database schemas (PostgreSQL) to efficiently store, index & query parsed vendor quotes and itemized transactional logs - Cloud Hosting & Serverless Architecture for deployment of scalable serverless backend webhooks and API endpoints on cloud (Node.js/FastAPI) using serverless cloud environment integrations with Meta WhatsApp Cloud API

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Replit – for all source code

Runpod- connect open source LLM, such as Deepseek V4.1 flash 1017

Meta developer whatsapp API key

Gemini

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied Design-Thinking to ensure **QuoteBot AI** is technically viable, user-friendly, and ready for Phase 2 deployment:

- **Empathize & Define:** Research with SME procurement teams revealed heavy reliance on WhatsApp for trade, creating major operational bottlenecks. We defined the core need: zero-friction quotation parsing without leaving WhatsApp.
- **Ideate & Prototype:** We built an in-chat AI assistant using multi-modal Vision-LLMs to process chats, voice notes, photos, and PDFs, automatically generating structured summaries and downloadable PDF reports.
- **Test & Iterate:** Tested against real-world supplier inputs—handwritten receipts, noisy chat logs, and multi-page price sheets—to optimize OCR precision and JSON extraction accuracy.

Phase 2 Deployment Strategy:

1. **Scalable Infrastructure:** Transition to containerized cloud microservices (FastAPI/Node.js) with auto-scaling LLM endpoints (vLLM/RunPod) and asynchronous Redis task queues for high-concurrency PDF rendering.
2. **Enterprise Security:** Enforce PostgreSQL Row-Level Security, end-to-end media encryption, and strict data-retention policies.
3. **System Integration:** Add CSV/Excel exports and Webhook/API sync for seamless integration with accounting software like QuickBooks.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess [Innovation and Creativity as well as Feasibility of build]. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s)	SDG 8 — Decent Work and Economic Growth
Metric Category	**Your Response**
Identify the exact UN SDG subtarget your project addresses *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: Target 8.3** — "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services." **Secondary: Target 8.2** — "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Over **80% of business-to-business transactions** across emerging SME trade markets currently depend on unstructured WhatsApp conversations without automated record-keeping.

Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Core Measurable KPI: % reduction in quotation processing time** (Targeting an 80% to 90% reduction, bringing manual parsing down from ~15 minutes per quote to under 30 seconds)
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	95% accuracy rate in structured quotation data extraction from unstructured chats, photos, and PDF files.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

√ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

√ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

√ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

√ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date] 22 October 2026